$$A \cap C = \{ (2,2), (2,4), (2,6) \}$$

$$n(A \cap C) = 3$$

$$P(A \cap C) = \frac{3}{36}$$

$$B \cap C = \{ (2,2), (4,2), (6,2) \}$$

$$n(B \cap C) = 3$$

$$P(B \cap C) = \frac{3}{36}$$

$$A \cap B \cap C = \{ (2,2) \}$$

$$n(A \cap B \cap C) = 1$$

$$P(A \cap B \cap C) = \frac{1}{36}$$

NOTE

If A and B are mutually exclusive events, $P(A \cap B) = 0$

The following relations may be obtained from these probabilities

$$P(A \cap B) = P(A) \times P(B)$$

$$\frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$$

$$P(A \cap C) = P(A) \times P(C)$$

$$\frac{3}{36} = \frac{1}{6} \times \frac{1}{2}$$

$$P(B \cap C) = P(B) \times P(C)$$

$$\frac{3}{36} = \frac{1}{6} \times \frac{1}{2}$$

$$P(A \cap B \cap C) \neq P(A) \times P(B) \times P(C).$$

$$\frac{1}{36} \neq \frac{1}{6} \times \frac{1}{6} \times \frac{1}{2}$$

NOTE

If A and B are independent events, $P(A \cap B) = P(A) \times P(B)$

The above relations show that when the events A, B and C are considered in pairs, they are independent. But, when all the three events are considered together, they are not independent.

8.7 Multiplication Theorem on Probability

Theorem 8.5: (Multiplication Theorem of Probability)

If A and B are any two events of an experiment, then

$$P(A \cap B) = \begin{cases} P(A)P(B \mid A), & \text{if } P(A) > 0 \\ P(B)P(A \mid B), & \text{if } P(B) > 0 \end{cases}$$